

RF power and spectrum limits — 5.8 GHz FMCW radar in Sweden

Research findings for the `antsdr-fmcw-radar` project. Researched 2026-08-09 against the regulations in force (PTSFS 2025:1, in force 17 March 2025) and the AD9361 datasheet. Regulations change — re-check PTS before acting on anything here more than a year from now.

Bottom line

- Measured/verified TX ceiling is ~6.5 dBm typ at 5.5 GHz on the bare AD9361, so +5 to +6.5 dBm at 5.8 GHz for the E200. The project's "~7 dBm" is the optimistic reading.
- $f_c = 5.8$ GHz was a good choice. A 50 MHz chirp there spans 5.775–5.825 GHz, which sits inside both Swedish license-free regimes at once — the non-specific SRD band (5.725–5.875 GHz, 25 mW e.i.r.p.) and the amateur band (5.65–5.85 GHz, 200 W at the antenna port with a certificate). No band change is needed.
- The project is currently transmitting 28 dB below its own license-free ceiling because of the `SDR_TX_GAIN_MAX_DB = -40` clamp. That is 5× in range, free, no licence, no purchase.
- For real power (a PA), the route is a Swedish HAREC amateur certificate. It raises the ceiling from 25 mW e.i.r.p. to 200 W at the antenna port and exempts homebrew gear from CE/RED.
- The practical ceiling on a monostatic 100 %-duty FMCW radar is TX/RX isolation, not the law and not the PA.

1. Verified TX power (AD9361)

From Table 1 of the AD9361 datasheet — typical values, single tone into a 50 Ω load, 0 dB attenuation. The nearest characterized band to 5.8 GHz is 5.5 GHz.

Spec	800 MHz	2.4 GHz	5.5 GHz
Maximum output power (typ)	8 dBm	7.5 dBm	6.5 dBm
TX carrier leakage @ 0 dB attenuation	−50 dBc	−50 dBc	−50 dBc
TX carrier leakage @ 40 dB attenuation	−32 dBc	−32 dBc	−30 dBc
RX noise figure (maximum RX gain)	—	—	3.8 dB
TX OIP3	23 dBm	19 dBm	17 dBm

Other figures worth having on hand:

- TX power control range: 90 dB in 0.25 dB steps. That is the variable attenuator — no trim pads are needed on the bench, only the fixed protection block.
- RF input absolute maximum: +2.5 dBm peak. TX monitor input: 9 dBm peak.
- TX LO covers 47 MHz – 6.0 GHz; RX LO 70 MHz – 6.0 GHz.

Three consequences for this project:

- 6.5 dBm is the bare chip number. The E200 adds balun, filter and switch losses, and 5.8 GHz is above the characterized 5.5 GHz point. Budget +5 to +6.5 dBm.
- Attenuating TX makes leakage relatively worse — carrier leakage degrades from −50 dBc wide open to −30 dBc at 40 dB attenuation. That energy lands in the close-in range bins that `CFAR_MASK_N` already masks, so it is handled, but it explains why leakage does not scale down 1:1 when TX gain is backed off.
- The +2.5 dBm absolute maximum is why the ≥ 30 dB pad discipline on any TX→RX cable path is not optional. `SDR_TX_GAIN_MAX_DB` is the only software guard between a mistyped gain and a dead front end.

2. The legal frame

In Sweden every transmitter needs a licence under lagen (2022:482) om elektronisk kommunikation, 3 kap., unless it falls under PTSFS 2025:1, the PTS exemption regulation. That document is the whole game: roughly 260 numbered paragraphs, each one band + application + power ceiling. A DIY radar is legal only if it fits one of those paragraphs, or if you hold an individual licence.

Cabled work radiates nothing and is therefore unregulated. Part F (TX → pads → RX over coax) needs no permission of any kind.

3. Route A — license-free SRD at 5.8 GHz

PTSFS 2025:1 § 204: 5.725–5.875 GHz, non-specific short-range devices, maximum 25 mW e.i.r.p. No certificate, no application, no fee. The 5.775–5.825 GHz chirp fits with 50 MHz to spare at both edges.

25 mW e.i.r.p. = 14 dBm e.i.r.p., measured at the antenna, so TX cable loss is free headroom rather than a cost.

The arithmetic for Part G (19 dBi sector)

Setting	hardwaregain	TX port	e.i.r.p. (19 dBi)	Margin to 14 dBm
Current clamp	−40 dB	≈ −33.5 dBm	≈ −14.5 dBm	28.5 dB unused
At the legal ceiling	≈ −12 dB	≈ −5.5 dBm	≈ 13.5 dBm	~0.5 dB
With ~2 dB TX cable loss	≈ −10 dB	≈ −3.5 dBm	≈ 13.5 dBm	~0.5 dB
Wide open (needs a licence)	0 dB	≈ 6.5 dBm	≈ 25.5 dBm	11.5 dB over

Free-space range scales as $P^{(1/4)}$, so what that headroom is worth:

TX port power	Δ vs current	Relative range
−33.5 dBm (current clamp)	baseline	×1
−5.5 dBm (SRD ceiling, 19 dBi)	+28 dB	×5.0
+6.5 dBm (wide open)	+40 dB	×10
+33 dBm (2 W PA, amateur route)	+66.5 dB	×46

These are free-space figures that ignore clutter competition, pointing loss and isolation — the honest ordering, not a promise of absolute range.

Caveat: RED applies on this route

Equipment operating under an SRD exemption must meet the essential requirements of the Radio Equipment Directive. A homebrew SDR radar has no conformity assessment behind it. This is a paper problem for private bench work, but it is a real distinction from Route B, where homebrew is explicitly exempt.

4. Route B — amateur certificate (HAREC)

PTSFS 2025:1 § 203: 5.65–5.85 GHz, amateur, maximum 200 W p.e.p. "tillförd antennsystemet" — power delivered to the antenna system. Antenna gain is not counted against this limit, unlike the e.i.r.p.-capped SRD route. That single difference is what makes the amateur route worth having.

Radar is a legal amateur activity

PTSFS 2025:1 defines amatörradiotrafik as "icke yrkesmässig radiotrafik för övning, kommunikation och tekniska undersökningar, bedriven i personligt radiotekniskt intresse och utan vinstsyfte" — non-professional radio traffic for training, communication and technical investigations, pursued out of personal radio-technical interest and without profit motive. Radar experiments are technical investigations. The binding word is non-commercial: hobby yes, contract work no.

Obligations that actually apply to a radar

- Call-sign identification. § 27 (whose paragraphs 3–6 are incorporated by every band section) requires the call sign to be transmitted, and requires automatic transmitters — which an unattended radar is — to be identifiable by a regularly transmitted call sign. Practically: interleave a CW or voice ID between CPIs.
- Secondary status. In the Swedish frequency plan, amateur is secondary at 5.65–5.85 GHz; radiolocation is primary and military use is allocated across 5.091–5.85 GHz. No protection from them, and no interference to them.
- No harmful interference generally: "de tekniska egenskaperna ska anpassas så att de inte stör användningen av andra radioanläggningar". With a PA this means a bandpass filter after it — the AD9361 has LO leakage and image products that must not be radiated.

CE / RED exemption

RED 2014/53/EU Annex I excludes radio equipment used by radio amateurs unless it is made available on the market. Kits assembled by amateurs, commercial equipment modified by and for amateurs, and equipment built by amateurs themselves are all explicitly "not made available on the market". A homebrew SDR radar needs no CE conformity assessment on this route.

Getting the certificate

- SSA (Föreningen Sveriges Sändareamatörer) runs the exams under delegation from PTS. Register at examen.ssa.se; practice exams are there too.
- HAREC standard = CEPT Recommendation T/R 61-02. Exam fee ≈ 200 kr, certificate issue ≈ 390 kr (2025 figures). Call sign is assigned with the certificate (SA-series).
- The entry-level certificate (instegscertifikat, new in 2025) is useless here. It covers only 7.0–7.2, 14.00–14.35, 21.00–21.45, 28.0–29.7, 50–52 and 144–146 MHz at 25 W, and it states "får endast användas med CE-märkta radiosändare" — CE-marked transmitters only. Homebrew SDR is excluded, and it has no microwave bands anyway.

Above 200 W

Apply to PTS at radiotillstand.pts.se for up to 1000 W p.e.p. Requires a valid HAREC certificate. Up to five transmitter sites per licence, issued per calendar year and auto-renewed. Fee 380 kr per transmitter per year (PTSFS 2025:3, effective 2026-01-01). Eligible bands include 432–438 MHz, 1240–1258 MHz, 5.65–5.85 GHz, 10.0–10.5 GHz and 24.00–24.25 GHz.

5. Route C — test licence (testtillstånd)

PTS issues time-limited licences for research, development, testing and demonstration. This is the path if the work ever becomes commercial (which disqualifies the amateur route by definition), or if a band outside the amateur allocations is needed. Case-by-case assessment, higher fees than the amateur licence.

6. Every option inside the E200's tuning range

Amateur bands (70 MHz - 6 GHz), license-free with a HAREC certificate

Band	Max power to antenna system	Width	Status in the Swedish plan
50–52 MHz	200 W p.e.p.	2 MHz	secondary
144–146 MHz	200 W p.e.p.	2 MHz	amateur primary
432–438 MHz	200 W p.e.p.	6 MHz	co-primary with radiolocation
1240–1300 MHz	200 W p.e.p.	60 MHz	secondary
2400–2450 MHz	100 mW p.e.p.	50 MHz	secondary (Swedish restriction)
5650–5850 MHz	200 W p.e.p.	200 MHz	secondary

Sweden has no 70 MHz (4 m) allocation — the exemption list jumps from 50–52 MHz to 144–146 MHz. Note also that 2.4 GHz amateur is capped at 100 mW in Sweden, which is unusually low and rules that band out

for any power work. Outside the E200's range, 10.0–10.5 GHz also allows 200 W and is the classic amateur radar band.

For 1240–1300 MHz: ECC Decision (25)01 constrains 1258–1300 MHz to protect Galileo E6, and the >200 W licence covers only 1240–1258 MHz. Keep below 1258 MHz there.

License-exempt radiodetermination — no certificate needed by anyone

Band	Limit
2400.0–2483.5 MHz (§ 187)	25 mW e.i.r.p.
10.25–10.28 / 10.35–10.38 / 10.51–10.58 GHz	500 mW e.i.r.p.; 5 W if antenna gain > 20 dBi
13.4–14.0 GHz	25 mW e.i.r.p.
24.00–24.25 GHz	100 mW e.i.r.p.
6.0–8.5 GHz (level probing, § 210)	7 dBm/50 MHz peak, –33 dBm/MHz mean e.i.r.p.

The 10.5 GHz entries are the most generous license-free radar allocations in the whole regulation — 5 W e.i.r.p. with a >20 dBi antenna — but they are above the AD9361's 6 GHz ceiling and would need an external upconverter.

7. Neighbours in 5.725–5.875 GHz

The 5.8 GHz band is shared, and the chirp at 5.775–5.825 GHz has company:

- § 205: 5.795–5.815 GHz, road and vehicle telemetry, 2 W e.i.r.p. This is CEN DSRC — toll gantries. The chirp straddles it dead centre. Legal on both routes, but do not point the radar at a toll gantry, and expect transponders in passing vehicles to appear in the RD map.
- RLAN / data transmission occupies 5.470–5.725 GHz, i.e. immediately below the band.
- § 206 / § 207: ITS at 5.855–5.925 GHz, immediately above. Keep the sweep clear of the upper edge, and remember that the amateur allocation itself stops at 5.85 GHz.
- Weather radar sits at 5.6–5.65 GHz, just below the amateur band's lower edge.

Interference received is likely in this band; interference caused is the real risk once a PA is in line.

8. Consequences for the project roadmap

Part G (antennas, no PA). The SRD budget is the constraint, and there is 28 dB of it going unused. `SDR_TX_GAIN_MAX_DB = -40` is a leftover from cable loopback, where it protected the RX front end; on antennas the equivalent legal clamp is about –12 dB (or –10 dB allowing for TX cable loss). Both the sector's gain and the cable loss belong in that calculation.

Part H (PA). A 2 W FPV PA is 33 dBm — 24 dB under the 200 W amateur ceiling, and illegal by 19 dB under the SRD ceiling with the 19 dBi sector. So the PA step requires either the amateur certificate or a Faraday cage. This matches what `TOD0.md` and `hardware/materials.md` already assume; the certificate is the cheaper of the two.

Part J (synthetic wideband). The synthetic bandwidth ceiling is legal, not technical:

Route	Usable band	Max B_syn	Fine range resolution
SRD	5.725–5.875 GHz	150 MHz	~1.0 m
Amateur	5.65–5.85 GHz (centre ≈ 5.75 GHz)	200 MHz	~0.75 m
Cable / digital loopback	unconstrained	330 MHz as planned	~0.45 m

The amateur route widens the on-air ceiling from the 150 MHz currently noted in `TOD0.md` to 200 MHz, at the cost of moving the hop centre down to about 5.75 GHz.

What does not work anywhere: the old 900 MHz carrier. 880–915 MHz is cellular uplink and 925–960 MHz is downlink; a 50 MHz chirp centred at 900 MHz spans both and would degrade service over a wide area. Fine

in the soft model and over coax, never radiated.

9. EMF exposure

SSMFS 2008:18 (Strålsäkerhetsmyndigheten's general advice, ICNIRP-based) sets a public reference level of 10 W/m² above 2 GHz. Far-field power density is $S = \text{e.i.r.p.} / (4\pi R^2)$, so the distance at which the reference level is reached:

Configuration	e.i.r.p.	Distance to 10 W/m ²
SRD ceiling, 25 mW	0.025 W	centimetres — irrelevant
2 W PA + 19 dBi sector	158 W	~1.1 m
2 W PA + 30 dBi dish	2.0 kW	~4.0 m
200 W + 19 dBi sector	15.8 kW	~11.2 m

Inside the far-field boundary ($2D^2/\lambda \approx 16$ m for a 0.65 m dish at 5.8 GHz) the far-field formula over-estimates the hazard, so these numbers are conservative. Dummy-load discipline until the antennas point away from people, and never stand in front of a dish under power.

10. The real ceiling: TX/RX isolation

FMCW is 100 % duty and constant envelope, so p.e.p. equals average power and the transmitter is always on while the receiver listens. With the AD9361's +2.5 dBm absolute maximum RF input, a 2 W (33 dBm) PA needs at least 31 dB of isolation simply to avoid destroying the front end, and far more than that before weak returns stop being buried under leakage.

Practical implications, in the order they bite:

1. Separate TX and RX antennas with real spacing and a septum — the Part G isolation experiments are the gate for everything in Part H.
2. Step power up in stages: 1 W, then 10 W. 10 W already buys ×3.8 range over the wide-open SDR.
3. A bandpass filter after the PA, and a harmonic low-pass, before anything radiates.
4. A continuously-rated dummy load and heatsinking — 100 % duty means a burst rating is a lie.

The law permits 200 W long before the hardware can usefully transmit it.

Sources

- AD9361 datasheet, Analog Devices — Table 1 (transmitter/receiver specifications) and absolute maximum ratings. <https://www.analog.com/media/en/technical-documentation/data-sheets/ad9361.pdf>
- Post- och telestyrelsens föreskrifter (PTSFS 2025:1) om undantag från tillståndsplikt för användning av vissa radiosändare, in force 2025-03-17. <https://pts.se/regelbibliotek/>
- Post- och telestyrelsens allmänna råd (PTSFS 2025:2) om den svenska frekvensplanen.
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- PTS e-tjänst, amatörradio upp till 1000 W p.e.p. <https://radiotillstand.pts.se/amateur/create>
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- SSA, Bli radioamatör / Examen. <https://www.ssa.se/amatorradio/bli-radioamator/>
- Directive 2014/53/EU (RED), Annex I. <https://eur-lex.europa.eu/eli/dir/2014/53/oj/eng>
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- SSMFS 2008:18, Strålsäkerhetsmyndighetens allmänna råd om begränsning av allmänhetens exponering för elektromagnetiska fält.